

# Souhaib Ben Farhat

## AI FOCUS — 9+ YEARS SENIOR FULLSTACK ENGINEER

LLMs · RAG · CAG · MCP · LangGraph · LangChain · CLI · Python · TypeScript · React · Node.js



LinkedIn



GitHub

[↗ Portfolio](#)

Senior fullstack engineer with **9+ years** building SaaS products across **React**, **TypeScript**, and **Python**, now focused on **AI products** — **agentic systems** and **LLM-backed features**. I take fuzzy, under-defined problems and ship the thing that actually matters: designing and delivering features **end to end**, from backend services and agent tool-execution to polished, production-grade frontend experiences — turning ambiguous product ideas into software people rely on.

## Selected Achievements

- Built the **tool-execution**, **multi-tenant isolation**, and **conversation-persistence** layers behind parcelLab's conversational, AI-built product surfaces.
- **Redesigned the in-product AI Copilot** into a reusable chat component system — **SSE streaming**, **document upload**, and **stats/charts display**.
- Reviewed **400+ pull requests** in a year, helped roughly halve CI runtime, and hosted internal **AI workshops**.
- Designed a from-scratch **embeddable widget architecture** (~90% test coverage), adopted as the team's server-rendered standard.

## Experience

FEB 2025 — PRESENT

[✦ ASK AI](#)

### parcelLab <sup>↗</sup> — Software Engineer · Fullstack & AI

TRUSTED BY  ·  Microsoft ·  BOSS ·  dyson ·  PUMA

*Enterprise post-purchase platform — order tracking, delivery notifications, and returns for global retail brands.*

- Built the **tool-execution and multi-tenant isolation layer** of parcelLab's AI agent platform — the internal-API tools the agents call, **account-isolated multi-tenant threads**, and **conversation persistence and history** — surfaced to the UI over **SSE streaming** and powering an **AI build-by-conversation** experience where users compose product UI by chatting with an agent.
- Led the **end-to-end redesign and modernization of the in-product AI assistant** (Copilot) — integrating the LLM chat backend and rebuilding the chat experience into a modern, reusable **UI/UX component system**.
- Built a **global search and command palette** — keyboard-first, application-wide navigation and quick actions that cut everyday navigation friction across the portal.
- **Defined the architecture for a new embeddable widget platform** — Shadow-DOM isolation, JSON-driven layouts and theming, custom **OIDC/PKCE** auth, and **~90% test coverage** — later adopted as the team's standard **server-rendered, AI-buildable** approach.
- Led the **portal-wide design system** — a WCAG-AA, system-preference **dark theme** and a unified data-table / page-header / skeleton-loading component system across a large React app — raising the visual and accessibility baseline every later feature starts from.
- Drove **perceived-performance** work — faster initial load, a navigation progress bar, and app-wide skeleton loading — that made the app feel instant.

JUL 2021 — JAN 2025

✦ ASK AI

- Owned **features end to end across UI, API, and CDN**, shipping **Python / Node backend services** to keep delivery unblocked — fullstack breadth, not just frontend.
- Ran **product-enablement sessions** and authored **feature documentation** — walking product and sales teams through new capabilities end to end so they could confidently position, demo, and sell them.
- **Reviewed 400+ pull requests** in a year — one of the most active reviewers on the main app — helped **roughly halve CI runtime**, built in-repo “skills” tooling for non-frontend engineers, and ran internal **AI workshops** across teams.

React · TypeScript · Python · AI/LLM agents · AWS / CloudFront · Shadow DOM · CI/CD

## KONUX ↗ — Software Engineer · Frontend & Platform

TRUSTED BY  **Deutsche Bahn** ·  **NetworkRail** ·  **adif** ·  **TRAFIKVERKET** ·  **INFRABEL**  
Right On Track

*Predictive-maintenance SaaS for rail infrastructure — IoT + ML monitoring of railway point machines, surfaced to rail operators through monitoring dashboards.*

- Owned the **point-machine monitoring dashboard** — KONUX's flagship product surface where rail operators monitor point-machine and track health — and built the **data-visualization UI** and visual language behind it for sensor and time-series data.
- Migrated charting from **D3.js to Highcharts** — standardizing how sensor time-series and point-machine telemetry are rendered across the product for faster delivery and a consistent charting experience.
- Architected the move from a monolithic frontend to a modular **NX monorepo** — a scalable platform foundation that enabled code-sharing across client apps and independent, parallel delivery of each product surface.
- Built a **shared UI component library and design system** spanning multiple client apps — a common UX foundation other apps built on, driving visual and interaction consistency across the product.
- Set up **internationalization with an automated translation pipeline** — i18n tooling that automates translations across the apps — and integrated analytics and monitoring instrumentation, as platform/devex tooling.
- Built an in-house **feature-flagging** solution on AWS AppConfig with serverless Lambdas for safe, decoupled release control, and contributed **Spring Boot** microservice APIs — reaching across frontend and platform.

React · TypeScript · Redux · NX monorepo · D3 / Highcharts · i18n automation · Java / Spring Boot · AWS (AppConfig / Lambda)

JAN 2019 — JUN 2021

✦ ASK AI

## Klarx ↗ — Fullstack Developer (React / Ruby on Rails)

TRUSTED BY **6,000+ construction firms**

*Construction-project and machine-rental management platform.*

- Built and maintained the management dashboard in React and developed the supporting Ruby on Rails APIs.

- Delivered the invoicing module (incoming/outgoing invoices, online payments) and rental-workflow automation; modernized the codebase and UI architecture for scale.

React · Redux · Ruby on Rails · PostgreSQL · REST

2017 — 2018

ASK AI

Earlier — Full-stack & mobile engineering

MOTIUS CLIENTS  BMW · BOSCH · SIEMENS

Motius (React / Django), Lynq (Android), and Xedyas (Node.js) — building web and mobile products across R&D and early-stage companies.

React · Django · Node.js · Android

Skills

Frontend

React   TypeScript   Redux / RxJS   Design systems   Shadow DOM  
Data-visualization (D3 / Highcharts)   Accessibility (WCAG)

AI / LLM

Anthropic / OpenAI APIs   LangChain / LangGraph   MCP (Model Context Protocol)  
RAG (Retrieval-Augmented Generation)   CAG (Cache-Augmented Generation)  
Tool-calling   SSE streaming

Backend

Python (Django-style REST)   API design   PostgreSQL   OIDC / PKCE   Node.js  
GraphQL   Rails   Spring Boot

Cloud / DevEx

AWS (Lambda, CloudFront)   CI/CD   Vitest / Jest / Cypress   NX monorepo

Leadership

Architecture & technical direction   Code review at scale   Mentoring  
Internal tooling & enablement   Workshops & documentation  
Product-enablement sessions   Cross-team initiatives

Education

ESPRIT — School of Engineering

Engineer's Degree, Computer Software Engineering  
2015 — 2018 · TUNIS

ESC — Higher School of Business

Research Master's, Information Systems & Decision Support  
2014 — 2016 · TUNIS

ESC — Higher School of Business

Bachelor's, Software Development  
2011 — 2014 · TUNIS

Languages

|         |        |
|---------|--------|
| English | C1     |
| German  | B2     |
| French  | C1     |
| Arabic  | Native |